

DNS-BASED AI TOOL USAGE & RISK REPORT

Detection of Generative-AI and LLM Service Access via DNS
Telemetry

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Source log: sample_dns_logs.csv

Classification: TLP:AMBER — Internal Distribution Only

This report summarizes generative-AI and large-language-model (LLM) service usage identified through passive DNS log analysis. Findings are intended to support shadow-AI discovery, data-exfiltration risk assessment, and acceptable-use-policy enforcement.

51	11	9	5
Flagged AI DNS Queries	High-Risk Queries	Distinct AI Services Seen	Users With AI Activity

1. Executive Summary

Automated parsing of `sample_dns_logs.csv` identified **51** DNS queries to known generative-AI, LLM, and AI-assisted tooling domains out of 65 total entries reviewed, generated by **5** distinct users across **9** distinct AI services. Of the flagged activity, **11** queries (22%) were classified as **high risk** — primarily generative media and direct API access patterns that can indicate bulk content generation or sensitive data leaving the corporate network through an unsanctioned channel.

The most frequently contacted service was **chat.openai.com** (11 queries), followed by **gemini.google.com** (8) and **api.openai.com** (7). Direct API endpoint access (`api.openai.com`) is notable because it typically reflects programmatic or scripted use — e.g., a custom integration, browser extension, or CLI tool — rather than casual browser-based chat use, and merits closer review for data-handling practices. High-risk generative-media services (**midjourney.com**, **elevenlabs.io**) accounted for 11 of the 51 flagged queries, concentrated among users **mchen**, **jdoe**, and **rpatel**.

One previously unclassified destination, **unknown-ai-tool.example**, generated 3 queries from users **mchen** and **rpatel** and is not yet mapped to a known AI provider in the detection ruleset. This domain is recommended for manual triage (Section 5).

2. Risk Breakdown

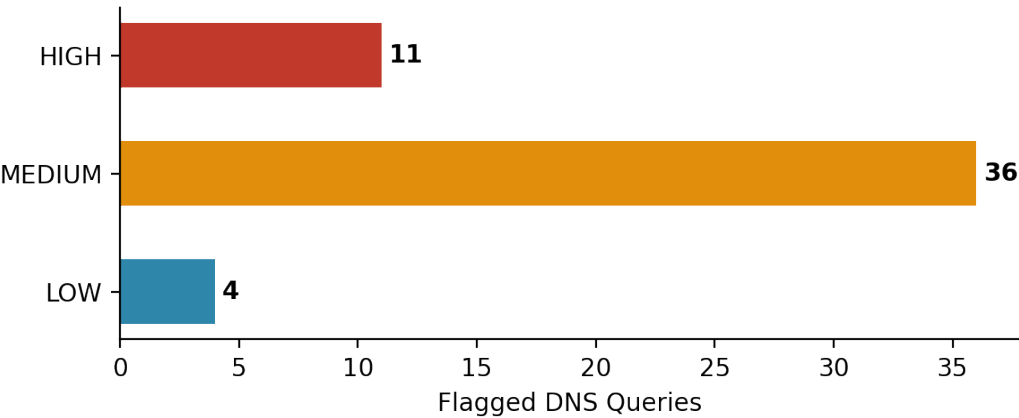


Figure 1. Flagged DNS queries by assigned risk level. Risk is determined by taking the higher of the ruleset-defined risk for the matched domain and the risk value recorded in the source log.

Risk Level	Queries	% of Flagged	Typical Indicator
HIGH	11	22%	Direct API access, image/video/audio generation — highest exfiltration & shadow-AI risk
MEDIUM	36	71%	Browser-based chat assistants, AI search — moderate risk, monitor for sensitive prompts
LOW	4	8%	Productivity tools with embedded AI features — lowest immediate risk

3. Top AI Services Detected

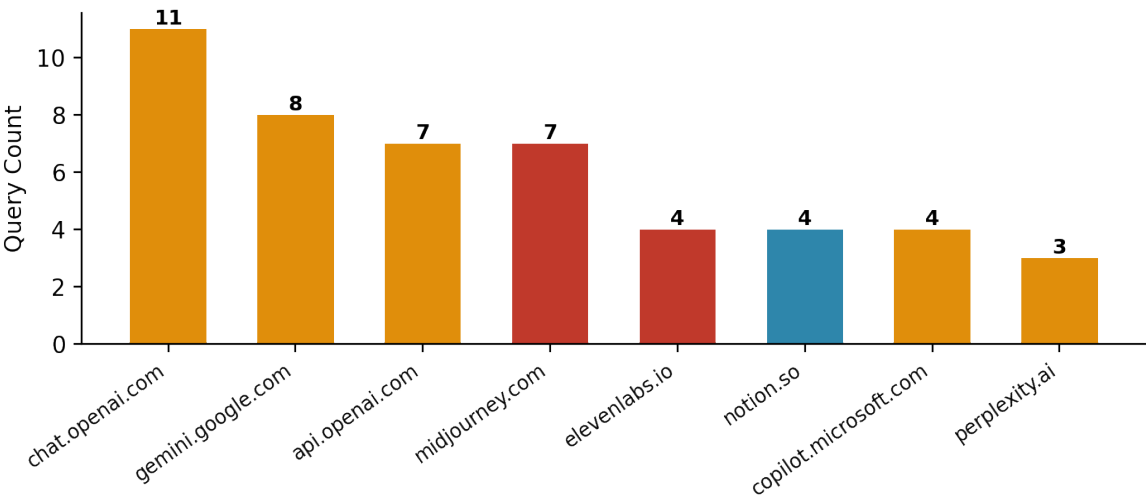


Figure 2. Query volume for the eight most-contacted AI domains. Bar color indicates risk level (red = high, orange = medium, blue = low).

Domain	Category	Risk	Queries
chat.openai.com	LLM Assistant	MEDIUM	11
gemini.google.com	LLM Assistant	MEDIUM	8
api.openai.com	LLM API	MEDIUM	7
midjourney.com	Image Generation	HIGH	7
elevenlabs.io	Audio Generation	HIGH	4
notion.so	AI-Powered Tool	LOW	4
copilot.microsoft.com	AI Assistant	MEDIUM	4
perplexity.ai	AI Search	MEDIUM	3
claude.ai	LLM Assistant	MEDIUM	3

4. User Activity Breakdown

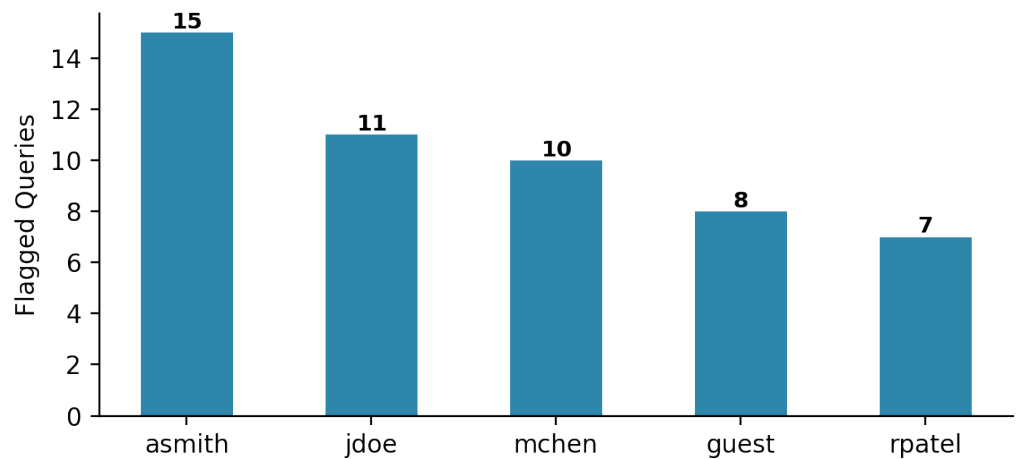


Figure 3. Total flagged AI-related DNS queries per user.

User	Unique AI Services	Total Queries	Highest-Risk Service Used
asmith	9	15	elevenlabs.io
jdoe	7	11	midjourney.com
mchen	7	10	midjourney.com
guest	6	8	midjourney.com
rpatel	4	7	midjourney.com

5. High-Risk Findings & Items for Triage

The following individual queries are flagged at **high** risk and warrant the closest review, along with one unclassified domain recommended for manual triage.

Timestamp (UTC)	User	Source IP	Domain	Category
2026-06-03 09:01:49	asmith	10.24.8.22	elevenlabs.io	Audio Generation
2026-06-03 09:05:46	mchen	10.24.9.31	midjourney.com	Image Generation
2026-06-03 09:12:04	guest	10.24.20.99	midjourney.com	Image Generation
2026-06-03 09:18:16	mchen	10.24.9.31	midjourney.com	Image Generation
2026-06-03 09:24:23	jdoe	10.24.8.15	midjourney.com	Image Generation
2026-06-03 09:25:51	jdoe	10.24.8.15	elevenlabs.io	Audio Generation
2026-06-03 09:29:03	jdoe	10.24.8.15	elevenlabs.io	Audio Generation
2026-06-03 09:29:50	rpatel	10.24.10.44	midjourney.com	Image Generation

Timestamp (UTC)	User	Source IP	Domain	Category
2026-06-03 09:36:19	rpatel	10.24.10.44	midjourney.com	Image Generation
2026-06-03 09:41:02	asmith	10.24.8.22	midjourney.com	Image Generation
2026-06-03 09:48:42	rpatel	10.24.10.44	elevenlabs.io	Audio Generation

Unclassified Domain Requiring Triage

unknown-ai-tool.example was queried 3 times by users **mchen** and **rpatel** and is logged with an **unknown** risk rating because it is not yet present in the detection ruleset. Recommend manual review of the destination IP and, if confirmed as an AI service, adding it to the custom-domains list (- - custom-domains) so future runs classify it automatically.

6. Recommendations

Review API-level access. api.openai.com traffic from guest, mchen, asmith, and jdoe accounts suggests scripted or integrated use. Confirm whether this traffic originates from an approved business integration and that no API keys or sensitive payloads are exposed outside sanctioned channels.

Clarify policy on generative-media tools. midjourney.com and elevenlabs.io together account for 11 high-risk queries across three users. Confirm these tools are covered by an acceptable-use or data-handling policy, particularly regarding upload of proprietary or customer content.

Triage the unclassified domain. Investigate unknown-ai-tool.example and update the detection ruleset (custom domain list) once its purpose is confirmed.

Re-evaluate the guest account. The guest account generated 8 flagged queries to AI assistants and search tools; confirm this aligns with guest network usage policy.

Schedule recurring detection runs. Re-run dns_ai_parser.py against ongoing DNS exports (e.g., daily or weekly) and track risk trends over time to identify emerging shadow-AI adoption.

Appendix A: Methodology

Detection was performed using **dns_ai_parser.py**, an internal tool that scans DNS query logs and flags entries matching a maintained ruleset of known AI/LLM/generative-AI service domains (suffix-matched, so subdomains are automatically captured). Each match is tagged with a category (e.g., LLM API, Image Generation, AI Search) and a risk level (low / medium / high). Where the source log already carries a risk annotation higher than the ruleset default, the higher value is kept.

This run parsed **sample_dns_logs.csv** with default settings (`--min-risk low --format both`), parsing 65 total log lines: 51 matched the AI ruleset and were flagged, 14 were non-AI traffic (e.g., `drive.google.com`, `github.com`) and were excluded, with zero parse errors.

Limitations: DNS-based detection identifies that a destination was resolved, not the content or volume of data exchanged; it cannot distinguish a single login from sustained bulk usage. Encrypted DNS (DoH/DoT) bypassing the monitored resolver, or use of IP addresses directly, will not appear in this dataset. Domains outside the current ruleset are not flagged unless added via the `custom-domains` option.

Output Files Generated

- `sample_dns_logs_flagged.csv` — full list of flagged entries
- `sample_dns_logs_domain_summary.csv` — per-domain rollup
- `sample_dns_logs_user_summary.csv` — per-user rollup
- `sample_dns_logs_report.json` — machine-readable combined report